Public Transportation Data Analysis & Dashboard Report

Project Overview

This project presents a comprehensive analysis of passenger and operational data from a public bus transportation system. The goal was to evaluate ridership patterns, operational efficiency, and user demographics to identify opportunities for system optimization. The entire project was conducted using **Microsoft Excel**, where a fully interactive dashboard was developed from scratch to visualize trends, track performance metrics, and support data-driven recommendations.

Objectives

- Analyze bus utilization to determine efficiency and demand across the network.
- Identify peak and off-peak travel hours to optimize scheduling.
- Examine rider demographics, including age, gender, and occupation.
- Assess route performance in terms of ridership and revenue.
- Provide strategic, actionable recommendations to enhance operations.

Data & Methodology

- Raw transportation data was imported and cleaned using Excel, including filtering, sorting, and handling missing or inconsistent values.
- Data modeling was performed using pivot tables to create meaningful groupings and aggregations.
- Interactive dashboards were developed using slicers, drop-down lists, charts, and conditional formatting for ease of analysis.
- Key performance indicators (KPIs) were calculated, such as total passengers, average riders per trip, utilization percentages, and revenue per route.

Key Findings

- **Utilization**: 36 buses were classified as over-utilized, while 19 were under-utilized, indicating a need for improved allocation.
- **Peak Hour**: The highest rider volume occurred at 11:41 AM, while 12:34 PM marked the off-peak period.

- **Time Analysis**: 64.61% of ridership occurred during PM hours compared to 35.39% in the AM.
- **Demographics**:
  - The majority of riders were between the ages of 30 and 49.
  - Professionals and self-employed individuals represented the largest occupation groups.
  - 74% of riders were male.
- **Route Performance**:
  - The East-West Express route had the highest ridership, while the South Line was the least used.
  - Average ridership per trip was 33 passengers.
- **Revenue**: Total revenue was recorded at $183,130.
- **Trend Analysis**:
  - Ridership in December reached 5,650, whereas January dropped significantly to 930.
  - Year-over-year ridership dropped by 83.5%, indicating a potential system or market issue.

Strategic Recommendations

1. **Bus Reallocation**: Reassign buses from under-utilized routes to over-utilized ones to balance demand and improve service coverage.
2. **Schedule Optimization**: Increase service frequency during peak hours to accommodate higher passenger loads.
3. **Demographic Targeting**: Develop targeted marketing campaigns focused on professional and middle-aged riders to boost engagement.
4. **Promotions by Day**: Leverage high-ridership days (e.g., Mondays) for fare promotions or loyalty programs.
5. **Ridership Recovery Plan**: Conduct surveys or customer feedback analysis to identify reasons behind the dramatic ridership decline and address them accordingly.

Tools and Techniques Used

- **Microsoft Excel** for:
  - Data Cleaning (sorting, filtering, missing value handling)
  - Pivot Tables and Charts
  - Conditional Formatting
  - Slicers and Drop-Downs for interactivity
- Data modeling and aggregation
- Dashboard design and layout optimization
- Trend and demographic analysis

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